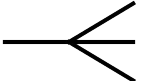


Crow's Foot Vocabulary (Symbols)

One = vertical bar |

Many = crow's foot 

Zero (optional) = circle O

These are the 'words' for building relationships.

Crow's Foot Phrases (Combinations)

$| \rightarrow \text{---} \swarrow \searrow \rightarrow$ one-to-many

$O \rightarrow \text{---} \swarrow \searrow \rightarrow$ zero-to-many

$| \text{---} \text{---} | \rightarrow$ exactly one-to-one

$O \text{---} \text{---} | \rightarrow$ zero-to-one

Symbols on both ends form a phrase.

Crow's Foot Sentences (Examples)

Let's diagram these:

Author |—>< Book → One author can write many books.

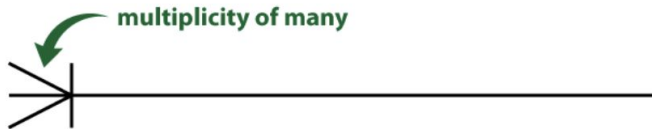
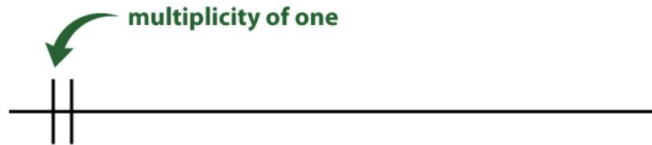
Book |—| ISBN → One book has exactly one ISBN.

Customer O—>< Order → A customer may have zero to many orders.

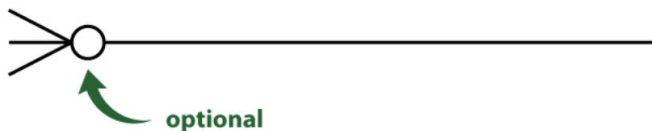
Crow's Foot Phrases (Cardinality)

Relationships have two indicators. These are shown on both sides of the line.

- The first one (often called multiplicity) refers to the *maximum* number of times that an instance of one entity can be associated with instances in the related entity. It can be one or many.

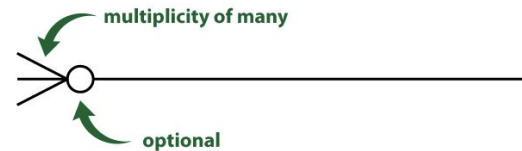


- The second describes the *minimum* number of times one instance can be related to others. It can be zero or one, and accordingly describes the relationship as optional or mandatory.



Finally, there are four possible edges to the relationship, illustrated here:

- zero or many



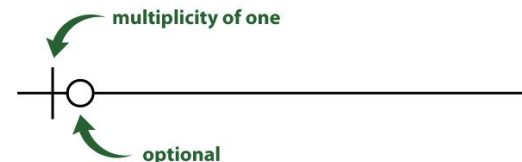
- one or many



- one and only one



- zero or one



Examples

Relationship degrees make them readable as :

- One-to-one



- One-to-many



- Many-to-many

